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## FILTER SYSTEM FOR FILTERING FLUIDS


### Abstract

A filter system for filtering fluids includes a housing, a fluid inlet formed in a first housing part, a fluid outlet formed in a second housing part, a flat non-serviceable filter element operably accommodated in the housing, wherein a fluid flow passes transversely through the filter element from the fluid inlet to the fluid outlet. The filter element has at least two filtration areas of different filtration densities including a first filter medium having a high filtration density and a second filter medium having a low filtration density, wherein the fluid flow passes the at least two filtration areas in parallel.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of priority of German Application DE 10 2024 106 158.8, filed on Mar. 4, 2024, in the German Patent and Trademark Office, the entire content of which is hereby incorporated by reference.

BACKGROUND

[0002] The invention relates to a filter system for filtering fluids, for example a liquid fluid filter system.

[0003] The filter system relates to filters for filtering a liquid fluid, for instance in cooling systems or in transmission systems in internal combustion engines, hybrid and battery electric vehicles and includes electric axle filter solution in battery electric vehicles and new energy vehicles. Such systems use suction side filters located in a tank such as, for instance, an oil pan or placed between the oil pan and the oil pump. Usually, the inlet is connected or placed inside the oil pan and the outlet connected to the oil pump. The purpose of these filters firstly is to prevent damage of the suction pump from dirt or abrasives in the transmission or coolant oil and secondly to prevent damage to the gear mechanisms from the dirt or abrasives. These filters can be used standalone or in conjunction with a pressure side filter placed after the pump in a transmission oil system. Existing suction side filters traditionally use either a strainer or filter medium placed inside the housing. The housing is used to hold the media in such a manner that one face of the media/strainer receives the unfiltered oil from the inlet and the filtered oil that emerges from the other face of the media/strainer flows to the outlet. The filtration media can be comprised of multiple layers of media that have different filtration properties, and the media can be arranged within the housing in multiple ways.

[0004] EP 1 994 974 B1 describes an oil filter device which has an oil inlet and an oil outlet, wherein a suction oil pump can be connected to the oil outlet, with which a negative pressure can be generated between the oil inlet and the oil outlet and has a coarse filter medium and a fine filter medium, which are arranged between the oil inlet and the oil outlet. The oil filter device is suitable for allowing a volume fraction of an oil penetrating through the oil inlet to pass permanently through the coarse filter medium and at the same time allowing the remaining volume fraction of the oil to pass through the fine filter medium, irrespective of the viscosity of the oil penetrating through the oil inlet. The fine filter medium is arranged upstream of the coarse filter medium in the flow direction. The fine filter medium is provided with at least one permanent opening through which oil can pass unfiltered. A clearance is provided between the fine filter medium and the coarse filter medium arranged behind it in the flow direction.

[0005] U.S. Pat. No. 10,876,443 B2 describes a filter device for filtration of oil, comprising a first filtration layer, a second filtration layer, and a frame element, wherein the frame element is arranged in a sandwich-like manner between the first filtration layer and the second filtration layer. The frame element has a multitude of passage orifices, such that a number of intermediate chambers corresponding to the multitude of passage orifices is formed between the first filtration layer and the second filtration layer. The frame element comprises an oil-permeable filter medium, by means of which filtration of the oil passing therethrough is effected.

SUMMARY

[0006] It is an object of the invention to provide an improved filter system for filtering fluids, for example a liquid fluid filter system.

[0007] According to an aspect of the invention the object is achieved by a filter system for filtering fluids, for example a liquid fluid filter system, comprising a housing, a fluid inlet formed in a first housing part, a fluid outlet formed in a second housing part, a flat non-serviceable filter element being operably accommodated in the housing, a fluid flow passing transversely through the filter element from the fluid inlet to the fluid outlet, the filter element having at least two filtration areas of different filtration densities, the fluid flow passing the at least two filtration areas in parallel.

[0008] Advantageous embodiments are described in the following description and the drawings.

[0009] According to an aspect of the invention a filter system for filtering fluids, for example a liquid fluid filter system, is proposed, comprising a housing, a fluid inlet formed in a first housing part, a fluid outlet formed in a second housing part, a flat non-serviceable filter element being operably accommodated in the housing, a fluid flow passing transversely through the filter element from the fluid inlet to the fluid outlet. The filter element has at least two filtration areas of different filtration densities, the fluid flow passing the at least two filtration areas in parallel.

[0010] The proposed filter system is a flat suction filter with non-pleated filter media for fluids like transmission oil and/or cooling medium such as oil or water glycole. In a first filtration area of the at least two filtration areas a filter medium or strainer with a high filtration density may be provided and in a second filtration area of the at least two filtration areas a filter medium or strainer with a low filtration density may be provided. Thus, part of the fluid may flow through the first filtration area which offers a higher pressure resistance and another part of the fluid may flow through the second filtration area with a lower pressure resistance. Thus, on an average, the differential pressure caused by the filter system is lower than when providing only a high filtration density to the fluid flow, and at the same time providing a sufficient filtration efficiency.

[0011] The filter medium with a low filtration density and the filter medium with a high filtration density can be arranged side by side establishing the two filtration areas.

[0012] Alternatively, the filter medium with a low filtration density and the filter medium with a high filtration density can be stacked one atop of the other in flow direction where the filter medium with a high filtration density has a cut-out exposing the filter medium with a low filtration density. The filter medium with a high filtration density can be arranged upstream of the filter medium with a low filtration density or vice versa.

[0013] Optionally, a filter medium with a low filtration density and a filter medium with a high filtration density can be arranged side by side establishing two filtration areas in a stack of filter media where a filter medium with a low filtration density and a filter medium with a high filtration density can be stacked one atop of the other in flow direction where the filter medium with a high filtration density has a cut-out exposing the filter medium with a low filtration density.

[0014] Advantageously, by using at least two filtration areas of different filtration densities, wherein the fluid flow passes the at least two filtration areas in parallel, the differential pressure caused by the suction filter for the fluid flow is reduced, where space and cost considerations prevent the use of a conventional mechanical bypass valve.

[0015] According to a favorable embodiment of the filter system, a first filtration area with a high filtration density may be larger than a second filtration area with a low filtration density. The first filtration area with the high filtration density may be larger, for example at least twice as large, or for example at least three times larger, or for example at least four times larger, than the second filtration area with the low filtration density. Thus, a sufficient filtration quality may be achieved by simultaneously reducing the differential pressure over the filter system in an advantageous way.

[0016] According to a favorable embodiment of the filter system, the filter element may at least comprise one first filter medium with the high filtration density and one second filter medium with the low filtration density. One first filter medium for fine filtration and one second filter medium for coarse filtration may advantageously be arranged in the housing of the filter system for achieving a first filtration area with a high filtration density and a second filtration area with a low

filtration density thus reducing the differential pressure for the fluid passing the filter element in total.
[0017] According to a favorable embodiment of the filter system, the first filter medium and the second filter medium may be arranged side by side transversely to the fluid flow. Thus, a first part of the fluid, entering the housing through the fluid inlet and passing through the first filter medium is fine filtered and a second part of the fluid, entering the housing through the fluid inlet and passing through the second filter medium is coarse filtered. Both parts of the fluid leave the housing after being filtered through the fluid outlet. The fluid inlet and/or fluid outlet can comprise several apertures.
[0018] According to a favorable embodiment of the filter system, a first divider may be arranged in the first housing part upstream of the filter element between the first filter medium and the second filter medium and a second divider may be arranged in the second housing part downstream of the filter element, the second divider being positioned opposite to the first divider relative to the filter element. The first divider and the second divider each may be provided with at least one flow channel for fluid passing between the first and the second filtration area of the filter element.
[0019] The first and second dividers offer a possibility for partitioning the fluid flow, for supporting and sealing the first and second filter medium against each other. The dividers are arranged at both sides of an interface between the two filtration areas.
[0020] Fluid that enters the housing through the fluid inlet in the first housing part has multiple paths to take. Firstly, there exists a fluid flow through the fine first filter medium and then to the fluid outlet in the second housing part. Else, the fluid can pass through one or more flow channels of the first divider, then through the coarse second filter medium with the low filtration density, next pass through the at least one flow channel of the second divider to the fluid outlet. When fluid flows through the filter system, the portion of fluid flow passing through the different filter media dynamically changes with variations in viscosity with the help of the first and second divider and flow resistance offered by the first and second filter media to the fluid flow.
[0021] The fluid flow passing through the at least one flow channel of the first and second divider to a major part may be streaming parallel to the filter element.
[0022] This embodiment offers a combination of divider mechanisms with the first and second filter medium that have different resistances to fluid flow. This combination allows for a dynamic change of amount of fluid flowing through the first and second filter medium when the fluid viscosity changes, for instance depending on a temperature of the fluid.
[0023] According to an embodiment of the filter system, the fluid inlet may be positioned in the first housing part and the fluid outlet may be positioned in the second housing part, for example opposing the first filtration area with the high filtration density. Thus, the major part of the fluid entering the first housing part may directly stream through the first filter medium, being fine filtered, and after passing the first filter medium streaming directly to the fluid outlet in the second housing part.
[0024] According to an embodiment of the filter system, the first divider may be arranged in the first housing part and/or the second divider may be arranged in the second housing part. Additionally or alternatively, the first divider may be an integral part of the first housing part and/or the second divider may be an integral part of the second housing part. The first divider may be fixed to the first housing part and the second divider may be fixed to the second housing part, e.g. by gluing or welding or the like for achieving a durable fixation to the first and second housing part.
[0025] According to an embodiment of the filter system, the first divider and/or the second divider may be provided with holding means for fixing the filter element, for example for clamping the filter element between the dividers. Thus, the filter element may be fixed and supported between the first and second divider. The first and second divider may even fix and tightly seal the interface between the first and second filter medium. The filter element for example can be clamped, welded,

glued or pressed between/to the dividers.

[0026] According to an embodiment of the filter system, the first filter medium and the second filter medium may be stacked atop of each other, wherein the first filter medium with a high filtration density may be provided with a cut-out for exposing the second filtration area with a low filtration density to the fluid flow. The cut-out may be located on a circumferential edge of the first filter medium so that the cut-out is only partially surrounded by the first filter medium with a high filtration density.

[0027] In this alternative embodiment a filtration media stack may be provided that has two or more distinct layers. In one embodiment, the layers may be arranged in a manner that the resistance to fluid flow decreases in the direction of the fluid flow. A portion of the filter media stack is cut-out, for example at the circumferential edge of the stack, in such a manner that if the fluid is to flow through this portion it would only flow through one or more filter media layers with lower resistance to fluid flow than in other areas of the filter element. Optionally, there can be one or more cut-outs positioned freely on the circumferential edges of the filter medium stack.

[0028] Fluid that enters the housing through the fluid inlet in the first housing part has multiple paths it can take. The fluid can flow through the complete filter media stack, then to the fluid outlet in the second housing part. Else, it can flow through the one or more cut-outs in the filter element, through one or more filter media layers with lower resistance to fluid flow and then to the fluid outlet. When fluid flows through the filter system, the portion of fluid flow passing through the filter media stack dynamically changes with changes in viscosity with the help of the one or more cut-outs. The at least one cut-out may be characterized by an absence of filter media having higher resistance to fluid flow.

[0029] The first filter medium with the high filtration density may be arranged upstream of the second filter medium with the low filtration density or, alternatively, the first filter medium with the high filtration density may be arranged downstream of the second filter medium with the low filtration density.

[0030] According to an embodiment of the filter system, ribs may be arranged on an inside of the first housing part and/or on an inside of the second housing part for directing the fluid flow and/or as a support of the filter element. Thus, the filter element may be supported mainly against pressure of the flowing fluid and/or to compensate for mechanical vibrations. The ribs may advantageously also serve for distributing and guiding the incoming or outgoing fluid over the surface of the filter element.

[0031] According to an embodiment of the filter system, a gap may be provided in flow direction between the ribs and the filter element. Thus, it may be ensured that the filter medium is not damaged by mechanical pressure by the ribs.

[0032] According to an embodiment of the filter system, recesses may be provided on an outer circumference of the first housing part and corresponding protrusions may be provided on an outer circumference of the second housing part or vice versa for positioning the first housing part and the second housing part relative to each other. A poka yoke positioning of the first housing part and the second housing part relative to each other can be achieved. Advantageously, mounting of the first housing part to the second housing part may be facilitated by the recesses and corresponding protrusions in the first and second housing part. In this way by the position of the fluid inlet upstream a desired filtration area of the filter element can be ensured. Likewise, by the position of the fluid outlet downstream a desired filtration area of the filter element can be ensured.

[0033] According to an embodiment of the filter system, an outer circumference of the filter element may be fixed to the first housing part and/or the second housing part, for example by gluing or welding or clamping. As the filter element may be configured as a non-serviceable filter element it may advantageously be fixed in a durable way to the first and/or second housing part, thus sealing the filter element to the housing as well as mechanically stabilizing against vibrations during operation.

[0034] According to an embodiment of the filter system, the first filter medium and/or the second filter medium may be of a single layer medium type. Alternatively, the first filter medium and/or the second filter medium may be of a multilayer medium type. Thus, the first and second filter media may comprise conventional non-pleated filter media or strainers used for filtering fluids like transmission oil or cooling liquid.

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## Description

BRIEF DESCRIPTION OF DRAWINGS

[0035] The present invention together with the above-mentioned and other objects and advantages may best be understood from the following detailed description of the embodiments in conjunction with the accompanying drawings, but not restricted to the exemplary embodiments shown in:

[0036] FIG. **1** an exploded view of a filter system for filtering fluids, for example a liquid fluid filter system, according to an embodiment of the invention;

[0037] FIG. **2** a detailed view of a divider of the filter system according to FIG. **1**;

[0038] FIG. **3** a sectional view of the filter system according to FIG. **1**;

[0039] FIG. **4** another sectional view of the filter system according to FIG. **1**;

[0040] FIG. **5** an exploded view of a filter system according to another embodiment of the invention; and

[0041] FIG. **6** a sectional view of the filter system according to FIG. **5**.

DETAILED DESCRIPTION

[0042] In the drawings, like elements are referred to with the same or similar reference numerals. The drawings are merely schematic representations, not intended to portray specific parameters of the invention. Moreover, the drawings are intended to depict only exemplary embodiments of the invention and therefore should not be considered as limiting the scope of the invention.

[0043] FIG. **1** depicts an exploded view of a filter system for filtering fluids, for example a liquid fluid filter system, according to an embodiment of the invention. The filter system **100** is a flat suction filter with non-pleated filter media **20**, **30** for fluids, such as transmission oil and/or cooling oil.

[0044] The filter system **100** comprises a housing **110** with a first housing part **112** and a second housing part **114**. A fluid inlet **102** is located in the first housing part **112**, and a fluid outlet **104** is located in the second housing part **114**. A flat non-serviceable filter element **10** is operably accommodated in the housing **110**, such that a fluid flow **50** can pass from the fluid inlet **102** through the filter element **10** to the fluid outlet **104**.

[0045] The filter element **10** has a flat body with two filtration areas **12**, **14** of different filtration densities arranged side by side in the flat body. The fluid flow **50** passes the two filtration areas **12**, **14** in parallel.

[0046] The first filtration area **12** with a high filtration density is larger than the second filtration area **14** with a low filtration density. The first filtration area **12** with the high filtration density may be at least twice as large, may be at least three times larger, or may be at least four times larger than the second filtration area **14** with the low filtration density. In the embodiment shown in FIG. **1** the first filtration area **12** is approximately three times larger than the second filtration area **14**.

[0047] In the first filtration area **12** the filter element **10** comprises a first filter medium **20** with the high filtration density and in the second filtration area **14** a second filter medium **30** with the low filtration density. The first filter medium **20** and/or the second filter medium **30** may be of single layer medium type or multilayer medium type. The first filter medium **20** and/or the second filter medium **30** can also be realized as strainers.

[0048] The first filter medium **20** and the second filter medium **30** are arranged side by side transversely to the fluid flow **50**.

[0049] A first divider **40** is arranged in the first housing part **112** upstream of the filter element **10** between the first filter medium **20** and the second filter medium **30** and a second divider **42** is arranged in the second housing part **114** downstream of the filter element **10**. The second divider **42** is hereby positioned opposite to the first divider **40**, relative to the filter element **10**. The dividers **40**, **42** are located beneath and above the interface between the first and second filter medium **20**, **30**, respectively.
[0050] The first divider **40**, which is shown in detail in FIG. **2**, is provided with a number of flow channels **44** parallel to the filter element **10**. The second divider **42** is provided with one broad flow channel **46**. The flow channels **44**, **46** let the fluid pass between the first and the second filtration area **12**, **14** of the filter element **10**. Besides the filtration densities of the filter media **20**, **40**, the respective cross sections of the areas of the flow channels **44**, **46** can be used for defining a pressure difference between the filtration areas **12**, **14** of the filter element **10**.
[0051] As seen from FIG. **1**, the fluid inlet **102** is positioned in the first housing part **112** and the fluid outlet **104** is positioned in the second housing part **114**, fluid inlet **102** as well as fluid outlet **104** opposing the first filtration area **12** with the high filtration density.
[0052] The first divider **40** may be arranged in the first housing part **112** and the second divider **42** may be arranged in the second housing part **114** and fixed to the housing parts **112**, **114** e.g. by gluing or welding. For this purpose, the first housing part **112** as well as the second housing part **114** are provided with recesses **138**, **140** to accommodate and fix the dividers **40**, **42** on the inside **116**, **118** of the first and second housing part **112**, **114**.
[0053] Alternatively, the first divider **40** may be an integral part of the first housing part **112** and/or the second divider **42** may be an integral part of the second housing part **114**.
[0054] The first divider **40** is provided with holding means **48** for fixing the filter element **10**, for example for clamping the filter element **10** between the dividers **40**, **42**. Thus, the filter element **10** may be fixed and supported between the first and second divider **40**, **42**. The first and second divider **40**, **42** may even fix and tightly seal the division line between the first and second filter medium **20**, **30**.
[0055] Fluid that enters the housing **110** through the fluid inlet **102** in the first housing part **112** has multiple paths to take. Firstly, there exists a fluid flow through the fine first filter medium **20** of the first filtration area **12** and then to the fluid outlet **104** in the second housing part **114**. Else, the fluid can pass through one or more flow channels **44** of the first divider **40**, then through the coarse second filter medium **30** in the second filtration area **14**, next pass through the flow channel **46** of the second divider **42** to the fluid outlet **104**.
[0056] Advantageously, by using the two filtration areas **12**, **14** of different filtration densities, wherein the fluid flow passes the two filtration areas **12**, **14** in parallel, the differential pressure that the suction filter offers to the flow of fluid is reduced.
[0057] In FIG. **2** a detailed view of the first divider **40** of the filter system **100** is depicted.
[0058] The number of flow channels **44** for passing the fluid between the first filtration area **12** and the second filtration area **14** may be recognized. The flow channels **44** are realized as indents in the body of the divider **40**. The holding means **48** on the upper side of the divider **40** are realized as two rails extending along a length of the divider.
[0059] A gasket **136**, e.g. an O-ring, may serve for sealing the fluid outlet **104** to a fluid duct or a fluid pump.
[0060] As may be seen from FIG. **1**, ribs **120** are arranged on an inside **116** of the first housing part **112** for directing the fluid flow **50** and/or as a support of the filter element **10**.
[0061] Ribs **122** are also arranged on an inside **118** the second housing part **114** but are not to be seen in FIG. **1**. Thus, the filter element **10** may be supported mainly against pressure of the flowing fluid and/or to compensate for mechanical vibrations or deformation. The ribs **120**, **122** may advantageously also serve for distributing and guiding the incoming or outgoing fluid over the surface of the filter element **10**.

[0062] Recesses **130** are provided on an outer circumference **128** of the first housing part **112** and corresponding protrusions **134** are provided on an outer circumference **132** of the second housing part **114** for positioning the first housing part **112** and the second housing part **114** relative to each, for example for poka yoke positioning the first housing part **112** and the second housing part **114** relative to each other.
[0063] Alternatively, recesses **130** may be provided in the second housing part **114** whereas corresponding protrusions **134** may be provided in the first housing part **112**.
[0064] FIG. **3** shows a sectional view of the filter system **100** in a section plane through the dividers **40**, **42**.
[0065] The dividers **40**, **42** are positioned in their corresponding recesses **138**, **140** of the first and second housing parts **112**, **114**. The filter element **10** is clamped between the dividers **40**, **42**, for example by means of the holding means, **48** of the first divider **40**, pressing the filter element **10** against the second divider **42**.
[0066] An outer circumference **16** of the filter element **10** is fixed to the first housing part **112** and the second housing part **114** and clamped between the first housing part **112** and the second housing part **114**. Additionally, the filter element **10** may be fixed by gluing or welding to the first and second housing part **112**, **114**.
[0067] The indents of the flow channels **44** of the first divider **40** are pressed against the inside **116** of the first housing part **112**. The indent of the flow channel **46** of the second divider **42** is pressed against the inside **118** of the second housing part **114**. Thus, the fluid may pass through the first flow channels **44** from the fluid inlet **102** to the second filtration area **14** and after being filtered in the second filtration area **14** back from the second filtration area **14** to the fluid outlet **104**.
[0068] Cooperation of protrusions **134** and corresponding recesses **130** for mounting the first and the second housing parts **112**, **114** is depicted at the outer circumference **128** of the housing **110**.
[0069] In FIG. **4** another sectional view of the filter system **100** in a section plane of 90° against the section plane of FIG. **3** is depicted. The fluid inlet **102** as part of the first housing part **112** is cut. The fluid outlet **104** is depicted in a side view.
[0070] In this view the first and the second dividers **40**, **42** are cut across.
[0071] The first filter medium **20** and the second filter medium **30** forming the filter element **10** in the first filtration area **12** and the second filtration area **14** are fixed on the outer circumference **16** of the filter element **10** by being clamped by the first and second housing parts **112**, **114** and/or additionally glued or welded to the first and/or second housing part **112**, **114**, and further by being clamped between the first and second divider **40**, **42**.
[0072] Further, the filter element **10** is supported by the ribs **120**, **122** on the inside **116**, **118** of the first and second housing part **112**, **114** across the surface of the filter element **10**.
[0073] Gaps **124**, **126** are provided in flow direction **50** between the ribs **120**, **122** and the filter element **10**. Thus, it may be ensured that the filter medium **20**, **30** is not damaged by mechanical pressure by the ribs **120**, **122**.
[0074] FIG. **5** depicts an exploded view of a filter system **100** according to another embodiment of the invention with an alternative configuration of the filter element **10**.
[0075] In this embodiment, the first filter medium **20** and the second filter medium **30** are stacked atop of each other. The first filter medium **20** with a high filtration density is provided with a cut-out **22** for exposing the second filtration area **14** with a low filtration density to the fluid flow. In this embodiment, the exposed surface of the second filter medium **30** is arranged downstream of the fluid flow **50**.
[0076] In this alternative embodiment the filtration media stack may be provided that has two distinct layers of first and second filter medium **20**, **30**. The layers are arranged in a manner that the resistance to fluid flow decreases in the direction of the fluid flow **50**. A portion of the filter media stack is cut-out, in such a manner that if the fluid is to flow through this portion it would only flow through one filter medium layer with lower resistance to fluid flow. There can be one or more cut-

outs **22** positioned freely on the circumferential edges **16** of the filter medium stack. The cut-out **22** is only partially surrounded by the first filter medium **20** with the high filtration density.
[0077] In the embodiment shown in FIG. **5**, the first filter medium **20** with the high filtration density is arranged upstream of the second filter medium **30** with the low filtration density. In an alternative embodiment the first filter medium **20** with the high filtration density may be arranged downstream of the second filter medium **30** with the low filtration density.
[0078] Fluid that enters the housing **110** through the fluid inlet **102** in the first housing part **112**, has multiple paths it can take. Firstly, the fluid flow through the complete filter media stack of first and second filter medium **20**, **30** in the first filtration area **12**, then to the fluid outlet **104** in the second housing part **114**. Else, it can flow through the cut-out **22** of the second filtration area **14**, through the second filter medium **30** with lower resistance to fluid flow and then to the fluid outlet **104**.
[0079] Advantageously, by using the two filtration areas **12**, **14** of different filtration densities, wherein the fluid flow passes the two filtration areas **12**, **14** in parallel, the differential pressure that the suction filter offers to the flow of fluid is reduced.
[0080] Recesses **130** are provided on an outer circumference **128** of the first housing part **112** and corresponding protrusions **134** are provided on an outer circumference **132** of the second housing part **114** for positioning the first housing part **112** and the second housing part **114** relative to each, for example for poka yoke positioning the first housing part **112** and the second housing part **114** relative to each other.
[0081] Alternatively, recesses **130** may be provided in the second housing part **114** whereas corresponding protrusions **134** may be provided in the first housing part **112**. Thus, the arrangement of ribs **120**, **122** on the inside **116**, **118** of the first and second housing part **112**, **114** is similar to the embodiment shown in FIGS. **1** to **4**, so that ribs **120** are arranged on an inside **116** of the first housing part **112** for directing the fluid flow **50** and/or as a support of the filter element **10** but are not seen in FIG. **5**.
[0082] Ribs **122** are also arranged on an inside **118** of the second housing part **114**,
[0083] Thus, the filter element **10** may be supported mainly against pressure of the flowing fluid and/or to compensate for mechanical vibrations or deformation. The ribs **120**, **122** may advantageously also serve for distributing and guiding the incoming or outgoing fluid over the surface of the filter element **10**.
[0084] FIG. **6** depicts a sectional view of the filter system **100** according to FIG. **5**.
[0085] The section plane is chosen as to cut through an area where the cut-out **22** of the first filter medium **20** is to be seen. In the first filtration area **12** the stack of both filter media **20**, **30** is to be seen, the fluid first passing the first filter medium **20** and then the second filter medium **30**. In the second filtration area **14** the fluid passed only the second filter medium **30**.
[0086] The arrangement of ribs **120**, **122** on the inside **116**, **118** of the first and second housing part **112**, **114** is supporting the filter element **10**.
REFERENCE NUMERALS
[0087] **10** filter element [0088] **12** first filtration area [0089] **14** second filtration area [0090] **16** outer circumference [0091] **20** first filter medium [0092] **22** cut-out [0093] **30** second filter medium [0094] **40** first divider [0095] **42** second divider [0096] **44** flow channel [0097] **46** flow channel [0098] **48** holding means [0099] **50** fluid flow [0100] **100** filter system [0101] **102** fluid inlet [0102] **104** fluid outlet [0103] **110** housing [0104] **112** first housing part [0105] **114** second housing part [0106] **116** inside of the first housing part [0107] **118** inside of the second housing part [0108] **120** rib [0109] **122** rib [0110] **124** gap [0111] **126** gap [0112] **128** outer circumference of the first housing part [0113] **130** recess [0114] **132** outer circumference of the second housing part [0115] **134** protrusion [0116] **136** gasket [0117] **138** recess [0118] **140** recess

## Claims

**1**. A filter system for filtering fluids, comprising: a housing, at least one fluid inlet formed in a first housing part, at least one fluid outlet formed in a second housing part, a filter element being operably accommodated in the housing, a fluid flow passing transversely through the filter element from the fluid inlet to the fluid outlet, the filter element having at least two filtration areas of different filtration densities, the fluid flow passing the at least two filtration areas in parallel.
**2**. The filter system according to claim 1, wherein a first filtration area with a high filtration density is larger than a second filtration area with a low filtration density, wherein the first filtration area with the high filtration density is at least twice as large, or at least three times larger, or at least four times larger, than the second filtration area with the low filtration density.
**3**. The filter system according to claim 1, wherein the filter element at least comprises a first filter medium with a high filtration density and a second filter medium with a low filtration density.
**4**. The filter system according to claim 1, wherein the first filter medium and the second filter medium are arranged side by side transversely to the fluid flow.
**5**. The filter system according to claim 4, wherein a first divider is arranged in a first housing part upstream of the filter element between a first filter medium and a second filter medium and a second divider is arranged in the second housing part downstream of the filter element, the second divider being positioned opposite to the first divider, relative to the filter element, wherein the first divider and the second divider each are provided with at least one flow channel for fluid passing between a first filtration area with a high filtration density and a second filtration area of the filter element.
**6**. The filter system according to claim 5, wherein the fluid inlet is positioned in the first housing part and the fluid outlet is positioned in the second housing part, opposing the first filtration area with the high filtration density.
**7**. The filter system according to claim 5, wherein the first divider is arranged in the first housing part and/or the second divider is arranged in the second housing part, and/or the first divider is an integral part of the first housing part and/or the second divider is an integral part of the second housing part.
**8**. The filter system according to claim 5, wherein the first divider and/or the second divider are provided with holding means for fixing the filter element between the first and second dividers.
**9**. The filter system according to claim 3, wherein the first filter medium and the second filter medium are stacked atop of each other, wherein the first filter medium with a high filtration density is provided with a cut-out for exposing the second filtration area with the low filtration density to the fluid flow.
**10**. The filter system according to claim 1, wherein ribs are arranged on an inside of the first housing part and/or on an inside of the second housing part for directing the fluid flow and/or as a support of the filter element.
**11**. The filter system according to claim 10, wherein a gap is provided in a flow direction between the ribs and the filter element.
**12**. The filter system according to claim 1, wherein recesses are provided on an outer circumference of the first housing part and corresponding protrusions are provided on an outer circumference of the second housing part for positioning the first housing part and the second housing part relative to each other.
**13**. The filter system according to claim 1, wherein an outer circumference of the filter element is fixed to the first housing part and/or the second housing part by gluing, welding, or clamping.
**14**. The filter system according to claim 3, wherein the first filter medium and/or the second filter medium are of single layer medium type or a multilayer medium type.